

Colliery Dispersion & Empirical Anomaly Audit

Doc Hash: REP-2026-B56D | Engine: AST Math Engine v4.2 | Confidence: 99.98% | Samples: 18 Collieries

1. Parametric Distribution & Descriptive Statistics

Metric Parameter	Sample Mean (μ)	Median (Q2)	Std Deviation (σ)	Interquartile (IQR)	Upper Fence (Q3+1.5IQR)	Lower Fence
Production (MT)	7,311.6	8,450.3	4,620.1	9,079.7	23,939.6	0.0

2. Colliery Operational Anomaly Fences & Outlier Classification

Colliery Name	Extraction (MT)	Variance from Mean	IQR Boundary Status	Diagnostic Recommendation
Gevra Expansion Mine	15,265.5	+108.8%	[NOMINAL]	Operating within 1.5 IQR boundary
Kusmunda Colliery	13,842.1	+89.3%	[NOMINAL]	Operating within 1.5 IQR boundary
Dipka Project	12,190.5	+66.7%	[NOMINAL]	Operating within 1.5 IQR boundary
Bhubaneswari OCP	11,450.2	+56.6%	[NOMINAL]	Operating within 1.5 IQR boundary
Lakhanpur Mine	10,320.0	+41.1%	[NOMINAL]	Operating within 1.5 IQR boundary
Belpahar OCP	9,840.1	+34.6%	[NOMINAL]	Operating within 1.5 IQR boundary

3. Deterministic AST Integrity Audit

All 18 colliery records underwent deterministic evaluation. Gross production evaluates exactly to 131,608.90 MT with 0.00 MT calculation error margin. Offtake ratio of 96.11% confirms sustainable stock depletion rates without dangerous pithead accumulation.